

DYNAMICAL SYSTEMS - MA3081

Solutions to Exercise Sheet 1

Exercise 1.1 (ODE Solving). Solve the following ordinary differential equations:

- a) $x''(t) - 2x'(t) + x(t) = 0$. *Hint: solve as linear ODE.*
- b) $tx'(t) - \sqrt{1 - x(t)^2} = 0$. *Hint: use separation of variables.*
- c) $x'(t) + x(t)\sin(t) = \cos(t)$. *Hint: use separation of variables and variation of constants.*

Solution.

- a) Since this ODE is linear, we make the ansatz $x(t) = e^{\lambda t}$. Inserting this ansatz into the ODE yields the characteristic polynomial $\chi(\lambda) = \lambda^2 - 2\lambda + 1$. By solving $\chi(\lambda) = 0$, it turns out that $\lambda_0 = 1$ is the only solution with multiplicity 2. Thus, a general solution of the ODE is given by an arbitrary linear combination of $e^{\lambda_0 t}$ and $te^{\lambda_0 t}$, i.e.,

$$x(t) = c_1 e^t + c_2 t e^t,$$

for some constants $c_1, c_2 \in \mathbb{R}$.

- b) We separate the variables to obtain

$$\frac{x'(t)}{\sqrt{1 - x(t)^2}} = \frac{1}{t}.$$

Integrating both sides on the time interval $[t_0, t]$ and conducting the substitution $x = x(t)$ on the left-hand side yields

$$\begin{aligned} \int_{x(t_0)}^{x(t)} \frac{1}{\sqrt{1 - x^2}} dx &= \int_{t_0}^t \frac{1}{s} ds \\ \arcsin(x(t)) &= \log(t) + c, \end{aligned}$$

where $c \in \mathbb{R}$ is a constant that depends on the initial condition $x(t_0)$. Solving for $x(t)$ results in

$$x(t) = \sin(\log(t) + c).$$

Beware: The problem is non-autonomous; hence, it is not invariant under translations in time. As such, we may not integrate on any time interval. In this case we should assume $t_0 > 0$ (or $t < 0$).

c) First, we consider the homogeneous problem

$$x'_{\text{hom}}(t) + x_{\text{hom}}(t) \sin(t) = 0.$$

Using separation of variables, as for the last ODE, we find

$$x_{\text{hom}}(t) = ce^{\cos(t)},$$

for a constant $c \in \mathbb{R}$. Now, we make the ansatz of variation of constants, i.e., $x(t) = c(t)e^{\cos(t)}$, for a time-dependent function $c(t)$. Its derivative is given by $x'(t) = c'(t)e^{\cos(t)} - c(t)e^{\cos(t)} \sin(t)$. Inserting the formulas for $x(t)$ and $x'(t)$ into the inhomogeneous ODE yields

$$\begin{aligned} c'(t)e^{\cos(t)} - c(t)e^{\cos(t)} \sin(t) + \sin(t)c(t)e^{\cos(t)} &= \cos(t), \\ c'(t)e^{\cos(t)} &= \cos(t), \\ c'(t) &= \frac{\cos(t)}{e^{\cos(t)}}, \\ c(t) &= c(0) + \int_0^t \frac{\cos(s)}{e^{\cos(s)}} ds. \end{aligned}$$

The final integral cannot be further simplified. Thus, a general solution of the inhomogeneous ODE is given by

$$x(t) = \left(d + \int_0^t \frac{\cos(s)}{e^{\cos(s)}} ds \right) e^{\cos(t)},$$

for a constant $d \in \mathbb{R}$.

□

Exercise 1.2 (Matrix Exponential).

- a) Consider the ODE $x'(t) = Ax(t)$ with initial condition $x(0) = x_0 \in \mathbb{R}^n$ for a matrix $A \in \mathbb{R}^{n \times n}$. Solve this ODE by computing the matrix exponential e^{At} of the following matrices:

$$\text{i) } A = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 \end{pmatrix};$$

$$\text{ii) } A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For which of these cases does this ODE induce a dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$ for $\mathcal{X} = \mathbb{R}^n$, $\mathcal{T} = \mathbb{R}$ and $\phi_t(x_0) = e^{At}x_0$?

- b) Now, let $A = A(t) \in \mathbb{R}^{n \times n}$ be time-dependent, i.e., we consider the ODE $x'(t) = A(t)x(t)$ with initial condition $x(0) = x_0 \in \mathbb{R}^n$. Under which assumptions is

$$x(t) = e^{\int_0^t A(s) ds} x_0$$

a solution formula? Under such conditions, prove that it indeed is one.

- c) Find an example of ODE in the form $x'(t) = A(t)x(t)$ such that for $\mathcal{X} = \mathbb{R}^n$, $\mathcal{T} = \mathbb{R}$ and $\phi_t(x_0)$ being the solution to the ODE at time $t \in \mathcal{T}$ with initial condition $x_0 \in \mathbb{R}^n$, the triple $(\mathcal{X}, \mathcal{T}, \phi_t)$ is not a dynamical system.

Solution.

- a) A solution to the ODE $x'(t) = Ax(t)$ with initial condition $x(0) = x_0$ is given by $x(t) = e^{At}x_0$. Thus, it remains to compute the matrix exponential:
- i) Note that the matrix A has a block form. The first block, $A_1 \in \mathbb{R}^{2 \times 2}$, is in Jordan normal form while the second, $A_2 \in \mathbb{R}^{2 \times 2}$, is symmetric. We can decompose $A_1 = D + N$ into a diagonal and a nilpotent matrix. Since D and N commute we have $e^{At} = e^{(D+N)t} = e^{Dt}e^{Nt}$. For $D = \text{diag}(d_1, \dots, d_n)$ the matrix exponential is $e^{Dt} = \text{diag}(e^{d_1 t}, \dots, e^{d_n t})$. For the nilpotent matrix N the defining series becomes a finite sum. For our choice of A one obtains

$$e^{A_1 t} = e^{Dt}e^{Nt} = \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{2t} \end{pmatrix} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} e^{2t} & te^{2t} \\ 0 & e^{2t} \end{pmatrix}.$$

For $A_2 = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ we find $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$ and hence

$$e^{A_2 t} = \frac{1}{2} \begin{pmatrix} e^{-t} + e^{3t} & -e^{-t} + e^{3t} \\ -e^{-t} + e^{3t} & e^{-t} + e^{3t} \end{pmatrix}.$$

In conclusion

$$e^{At} = \begin{pmatrix} e^{2t} & te^{2t} & 0 & 0 \\ 0 & e^{2t} & 0 & 0 \\ 0 & 0 & \frac{e^{-t} + e^{3t}}{2} & \frac{-e^{-t} + e^{3t}}{2} \\ 0 & 0 & \frac{-e^{-t} + e^{3t}}{2} & \frac{e^{-t} + e^{3t}}{2} \end{pmatrix}.$$

- ii) There are two possible ways:

1. Use complex diagonalization and proceed as in i) (for A_2).
2. Observe that $A^4 = \text{Id}$, so that the powers of A become periodic:

$$A^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, A^1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, A^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, A^3 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, A^4 = A^0, \dots$$

Therewith we obtain a power series for each component of $e^{At} = \sum_{k=0}^{\infty} \frac{A^k t^k}{k!}$, e.g. for the first entry we get $1 - \frac{1}{2}t^2 + \frac{1}{24}t^4 - \dots = \cos(t)$.

In any case we get $e^{At} = \begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix}$

As shown in the lecture, this ODE induces a dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$ with $\mathcal{X} = \mathbb{R}^n$, $\mathcal{T} = \mathbb{R}$ and $\phi_t(x) = e^{At}x$.

- b) The solution of $x'(t) = A(t)x(t)$ with $x(0) = x_0$ is given by

$$x(t) = e^{\int_0^t A(s) ds} x_0, \tag{1.1}$$

whenever $A(t)$ commutes with $A(s)$ for all $t, s \in \mathbb{R}$.

To prove that, we first define $M(t) := \int_0^t A(s)ds$. Then, $A(t)$ commutes with $M(t)$, due to the linearity of the integral. Consequently, $A(t)$ commutes with $M(t)^k$ for $k \in \mathbb{N}_0$. In general, the function $f(t) = M(t)^k$ has derivative

$$f'(t) = \sum_{\ell=1}^k M(t)^{\ell-1} M'(t) M(t)^{k-\ell},$$

for $k \geq 1$ and $f'(t) = 0$ for $k = 0$, by the product rule. Since $M'(t) = A(t)$ commutes with $M(t)$, this simplifies to

$$f'(t) = kA(t)M(t)^{k-1}.$$

Using this fact, we calculate the derivative of $x(t)$, that is given by (1.1):

$$\begin{aligned} x'(t) &= \frac{d}{dt} e^{M(t)} x_0 = \sum_{k=0}^{\infty} \frac{1}{k!} \left(\frac{d}{dt} M(t)^k \right) x_0 = \sum_{k=1}^{\infty} \frac{1}{k!} kA(t)M(t)^{k-1} x_0 \\ &= A(t) \sum_{k=1}^{\infty} \frac{1}{(k-1)!} M(t)^{k-1} x_0 = A(t) \sum_{k=0}^{\infty} \frac{1}{k!} M(t)^k x_0 \\ &= A(t) e^{M(t)} x_0 = A(t) x(t). \end{aligned}$$

Note that we can switch series and $A(t)$ under standard regularity assumptions, such as continuity of A .

- c) In general, this ODE does not induce a dynamical system, even when $A(t)$ commutes with $A(s)$, as already a one-dimensional example shows:

$$A(t) := \begin{cases} 0 & \text{for } t \leq 1 \\ 1 & \text{for } t > 1. \end{cases}$$

In this example, $\phi_1(x) = x$ and $\phi_2(x) = e^1 x$. However $\phi_1 \circ \phi_1(x) = x \neq e^1 x = \phi_2(x)$.

□

Exercise 1.3 (Dynamical Systems).

Decide whether the following triplets $(\mathcal{X}, \mathcal{T}, \phi_t)$ are dynamical systems or not.

- a) $\mathcal{X} = \mathbb{R}^2$, $\mathcal{T} = \mathbb{R}$, $\phi_t(x) = \begin{pmatrix} \sin(\lambda t) & -\cos(\lambda t) \\ \cos(\lambda t) & \sin(\lambda t) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ with some $\lambda \in \mathbb{R}$;
- b) $\mathcal{X} = \mathbb{R}$, $\mathcal{T} = \mathbb{Z}$, $\phi_t(x) = a^t x + b \frac{a^t - 1}{a - 1}$ with some $a \in \mathbb{R} \setminus \{1\}, b \in \mathbb{R}$.

Solution.

- a) No, since $\phi_0(x) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \neq \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$.
- b) Yes. Again $\phi_0(x) = x$ is easily checked.

Further

$$\phi_t \circ \phi_s(x) = a^t \left[a^s x + b \frac{a^s - 1}{a - 1} \right] + b \frac{a^t - 1}{a - 1} = a^{t+s} x + b \frac{a^{t+s} - 1}{a - 1} = \phi_{t+s}(x).$$

□

Exercise 1.4 (Invariant Sets).

- a) Let $(\mathcal{X}, \mathcal{T}, \phi_t)$ be an invertible dynamical system and $S \subset \mathcal{X}$ a positively (negatively) invariant set. Show that its complement $C = \mathcal{X} \setminus S$ is negatively (positively) invariant.
- b) Let $A \in \mathbb{R}^{n \times n}$. Show that the generalized eigenspaces of the matrix A are invariant under the flow generated by $x' = Ax$.
- c) Consider the following system of differential equations

$$\begin{aligned} x' &= y \\ y' &= x + \cos(x) \end{aligned} \tag{1.2}$$

as well as the function

$$E : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad E(x, y) = \frac{1}{2}y^2 - \frac{1}{2}x^2 - \sin(x).$$

Show that the level sets $S_c = \{(x, y) \in \mathbb{R}^2 \mid E(x, y) = c\}$ of the function E are invariant under the flow of (1.2).

Solution.

- a) Let S be positively invariant and assume that $\mathcal{X} \setminus S$ is not negatively invariant. That means there is some time $\tau < 0$ and some $x \in \mathcal{X} \setminus S$, such that $\phi_\tau(x)$ does not lie in $\mathcal{X} \setminus S$ consequently in S . Since S is positively invariant and $-\tau > 0$ we have $\phi_{-\tau}(\phi_\tau(x)) = x \in S$. But this is a contradiction to our assumption of $x \in \mathcal{X} \setminus S$. Hence $\mathcal{X} \setminus S$ is negatively invariant. For a negatively invariant S the reasoning is analogous.
- b) We know that the flow is explicitly given by $\phi_t(x_0) = e^{At}x_0 = \sum_{k=0}^{\infty} \frac{A^k t^k}{k!} x_0$. Let $v \in E$ be arbitrary for some (generalized) eigenspace E . Note that $Av \in E$ and inductively $A^k v \in E$ for any $k \geq 1$. For any $t \in \mathbb{R}$ we have $\phi_t(v) = \sum_{k=0}^{\infty} \frac{A^k v t^k}{k!}$. Since E is a linear subspace each of the partial sums of this series lies within E and since E is closed (like any linear subspace in finite dimensions) also the limit of the partial sums, i.e. the series, lies in E .
- c) We show that E is constant on every trajectory of the ODE. Let us take some trajectory $(x(t), y(t))$. We now compute

$$\begin{aligned} \frac{d}{dt} E(x(t), y(t)) &= \partial_x E(x(t), y(t)) \cdot x'(t) + \partial_y E(x(t), y(t)) \cdot y'(t) = \\ &= (-x - \cos(x)) y + y (x + \cos(x)) = 0, \end{aligned}$$

which shows that $E(x(t), y(t))$ is constant over time.

Consequently the level sets of E are invariant.

□